

First Semester Subjects Roadmaps

1. Discrete Mathematics (Term Grade)

- **Exams:** There will be two exams, midterm and endterm, each worth 50 marks.
- **Passing Requirement:** At least 18 points are required in each test to pass.
- **Midterm Topics:** Sets, Complex Numbers, Divisibility, Polynomials.
- **Endterm Topics:** Combinatorics, Vector Spaces, Matrices, Linear Equations.
- **Exam Pattern:** The exam will be similar to the sample quiz uploaded on e-learning.
- **Extra Points:** You can earn 1 extra point for each quiz solved within the given time.
- **Preparation Tip:** Solve the exercise sheet as it may help in retakes.

2. Introduction to Programming (Term Grade)

- **Exams:** There will be midterm and endterm exams.
- **Midterm Exam Parts:**
 1. **Written Part:**
 - Questions on number system conversions (e.g., binary to decimal).
 - Calculators are not allowed; practice fast calculations.
 2. **Theoretical Programming Part:**
 - Code will be provided; you will need to determine the output and explain loops.
- **Computer Part of Midterm:**
 - There will be 4 basic programming questions.
 - Practice questions from provided samples and focus on slides for understanding.
- **Endterm Exam:**
 - No written part; only computer part with 4 programming questions (lengthier).
 - Topics: Functions, Recursions, Dictionaries, Command Line Arguments, Reading from a file.

3. Computer Aided Mathematics and Visualization (CAMV) (Term Grade)

- **Exams:** There will be midterm and endterm exams.
- **Teaching Methods:**
 - Some professors teach MATLAB for both midterm and endterm.
 - Others teach GeoGebra for midterm and MATLAB for endterm.
- **Exam Pattern:**
 - The teacher will upload a guess paper or sample paper one week before the exam.
 - I uploaded the guess paper on my drive so you can prepare in advance.
- **Theoretical Knowledge:**

- Basic knowledge about equations, surfaces, etc., is needed.
- **Class Engagement:**
 - Be active in class and ask the teacher about solving questions as online lectures may differ.

4. Logics in Computer Science (Final Graded)

- **Exams:** No midterm and endterm exams.
- **Quizzes:**
 - Four quizzes during lab classes; one extra quiz for professional week attendees.
 - Earn extra points for every quiz.
- **Final Exam:** Questions will be similar to lab quizzes.
- **Preparation Materials:**
 - Teacher will upload sample quizzes with solutions.
 - I uploaded quiz papers with multiple versions and past papers for practice.
 - Lab slides are recommended for definitions.

5. Operating System (Final Graded)

- **Exams:** Midterm and endterm exams required for signatures.
- **Passing Requirement:** 50% required in each exam to pass and get a signature.
- **Lab Exams:** 25 MCQs sourced from Cisco.
 - A website will be shared for the questions.
- **Final Exam Preparation:**
 - Multiple past papers uploaded; final will be exactly the same.
 - PDF with solutions shared for reference.
 - Practice based on past papers is encouraged.